YOLO26-Mamba3: Selective State Space With Complex State Space for Dense Railway Pedestrian Detection

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***Abstract*—Dense pedestrian detection in railway scenes remains challenging because of frequent crowd gatherings and severe occlusion. Existing detectors in the You Only Look Once (YOLO) family often suffer from missed detections and false positives when pedestrians are tightly clustered. To address this issue, we propose YOLO26-Mamba3, a hybrid architecture that integrates YOLO26 with the Mamba3 Selective State Space Model (SSM) and its complex state representation. The method retains the efficient CNN backbone and NMS-free end-to-end inference pipeline of YOLO26 while replacing selected C2f modules in the neck network with sequence-aware Mamba3 modules. Specifically, the Mamba3 module first serializes the two-dimensional feature map into a one-dimensional sequence, uses the Selective SSM to focus on more informative pedestrian features under occlusion, and further employs the complex state space to encode relative spatial phase differences between adjacent individuals. This design helps distinguish closely adjacent pedestrians that are easily merged by conventional CNN-based detection heads. Experimental results on the self-built Dense Railway Pedestrian Detection Dataset show that YOLO26-Mamba3 improves mAP and recall under highly occluded conditions over the baseline YOLO26.**

***Index Terms*—Dense railway pedestrian detection, YOLO26, Mamba3, selective state space model, complex state space.**

I. Introduction

In RECENT years, deep learning-based object detection techniques, particularly those in the YOLO series [1], have been widely adopted in various visual detection tasks because of their excellent balance between speed and accuracy [2]. From YOLOv5 to the latest YOLO26 [3], [4], this family of detectors has achieved significant progress in generic object detection through anchor-free designs, decoupled heads, and end-to-end inference without Non-Maximum Suppression (NMS). However, in railway pedestrian detection, existing YOLO detectors still face a critical challenge: dense crowds and mutual occlusion [5], [6]. When multiple pedestrians walk in close proximity or partially overlzap, the limited receptive field of Convolutional Neural Networks (CNN) struggles to capture sufficient global context, causing the model to merge multiple individuals into a single detection box and resulting in numerous missed detections and false positives. This problem is particularly pronounced in scenarios such as stations during peak hours and level crossings [7].

To alleviate the dense-occlusion problem, recent studies have improved robustness in crowded or occluded scenes through multispectral pedestrian detection, multiview human modeling, and explicit occlusion-aware feature learning [8], [9], [10]. However, these methods still exhibit limitations in dense railway scenarios. Approaches designed for multispectral or multiview settings often require additional sensing devices or specific camera configurations, while occlusion-aware detector heads still inherit the locality bias of convolutional backbones [6], [11]. Transformer-based detectors provide strong global modeling capability, but their quadratic computational complexity, $O(N^2)$, results in excessive inference overhead, making them difficult to deploy on railway edge devices [7].

Recently, State Space Models (SSMs) have gained significant attention because of their linear complexity sequence modeling capability [12]. Among them, Mamba introduces the selective SSM [13], while Mamba-2 further improves efficiency through structured state space duality [14]. In computer vision, VMamba demonstrates that selective state space scanning can be effectively adapted to visual perception tasks [15]. The newly proposed Mamba3 further introduces a complex-valued state update rule for sequence modeling in dense scenes [16].

Inspired by these advances, we propose a hybrid architecture named YOLO26-Mamba3, which combines the efficient CNN-based feature extraction of YOLO26 with the sequential global modeling and complex state representation of Mamba3 [16]. Specifically, we retain the CNN backbone of YOLO26 and its NMS-free end-to-end inference pipeline, while designing a novel C2f_Mamba module to replace selected C2f modules in the neck network. This module adopts a dual-path structure with snake-scan positional encoding and a phase-preserving projection. It first serializes the two-dimensional feature map, then leverages the selective SSM to dynamically focus on the most discriminative pedestrian features under occlusion [17], and employs the complex state space to encode spatial phase differences between adjacent pedestrians. thereby effectively distinguishing closely spaced individuals in dense railway scenes [5], [6].

The main contributions are summarized as follows:

- We propose the YOLO26-Mamba3 architecture, integrating selective state space and complex state space to achieve complementarity between CNN and SSM.

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The source code is available at https://github.com/xlllchen/Yolo_Mamba.

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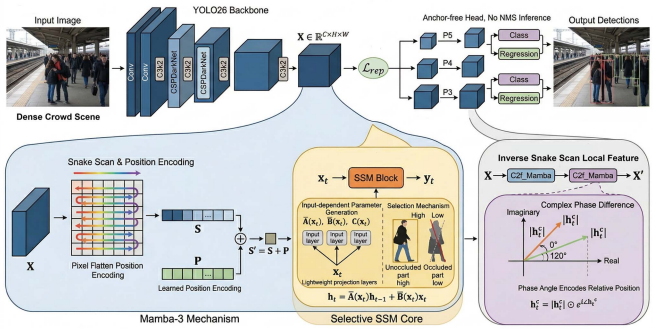


Fig. 1. Overall architecture of YOLO26-Mamba3 for dense railway pedestrian detection. The framework combines CNN-based local feature extraction with Mamba3-based sequence modeling to improve pedestrian discrimination in occluded railway scenes.

- We design a serialized feature processing method that introduces complex state phase difference encoding to enhance occlusion robustness and solve the merging problem of highly overlapping pedestrians.
- We validate the effectiveness of our method on our self-built Dense Railway Pedestrian Detection Dataset.

The remainder of this paper is organized as follows. Section II presents the design principles and implementation details of the YOLO26-Mamba3 architecture [18], including the overall architecture, the core mechanisms of the Mamba3 module, the design of the C2f_Mamba module, the hybrid neck-network design, and the loss function. Section III reports the experimental settings, results, and analysis. Section IV concludes the paper.

II. Method

A. Overall Architecture

YOLO26-Mamba3 is a hybrid architecture designed for dense pedestrian detection in railway scenes. As shown in Fig. 1, the model retains the CNN backbone of YOLO26 for local feature extraction while replacing the low-resolution C2f modules in the neck network with the proposed C2f_Mamba module, which integrates the selective state space modeling capability of Mamba3 with complex state representation [15]. The overall pipeline consists of feature serialization and positional encoding, sequence processing via the C2f_Mamba module, feature reconstruction, and output generation by an anchor-free, NMS-free detection head.

B. Core Mechanisms of the Mamba3 Module

1) Feature Serialization and Positional Encoding: Given a feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ from a certain layer of the neck network, Snake Scan is employed to flatten it into a one-dimensional sequence $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_L]$, where $L = H \times W$, while preserving spatial adjacency. A learnable positional encoding $\mathbf{P} \in \mathbb{R}^{L\times C}$ is subsequently added:

$$\mathbf{S}' = \mathbf{S} + \mathbf{P}. \tag{1}$$

The learnable positional encoding $\mathbf{P}$ is implemented as a trainable matrix of the same shape as the serialized sequence: $\mathbf{P} \in \mathbb{R}^{L\times C}$. Each position index i (from 1 to L) is associated with a C-dimensional learnable vector. The encoding is added directly to the serialized features after the snake-scan operation, so that the model can learn to preserve both absolute and relative order information along the scanning path.

2) Selective State Space Model: The core of Mamba3 is a discretized selective SSM, with the recurrence:

$$\begin{aligned}\mathbf{h}_t &= \overline{\mathbf{A}}(\mathbf{x}_t)\mathbf{h}_{t-1} + \overline{\mathbf{B}}(\mathbf{x}_t)\mathbf{x}_t,\\ \mathbf{y}_t &= \mathbf{C}(\mathbf{x}_t)\mathbf{h}_t.\end{aligned} \tag{2}$$

Unlike traditional SSMs that use fixed state transition matrices, Mamba3 computes the discrete parameters $\overline{\mathbf{B}}$ and $\mathbf{C}$ (and implicitly $\overline{\mathbf{A}}$ via input-dependent Δ) based on the current input $\mathbf{x}_t$; this is known as the selective SSM [13]. In dense railway pedestrian scenes, this selective mechanism provides the model with data-dependent contextual filtering capability, enabling it to focus on visible parts of unoccluded pedestrians while suppressing interference from occluded regions [6].

3) Complex State Space: To explicitly encode the relative spatial relationships between closely adjacent pedestrians, the hidden state is extended to the complex domain. Let the complex state $\mathbf{h}_t^c \in \mathbb{C}^d$ be represented in polar coordinates as follows:

$$\mathbf{h}_t^c = |\mathbf{h}_t^c| \odot e^{i\angle\mathbf{h}_t^c}, \tag{3}$$

where the modulus $|\mathbf{h}_t^c|$ encodes the absolute intensity of features, and the phase $\angle\mathbf{h}_t^c$ encodes the topological information of relative positions. For two severely occluded pedestrians, denoted as A and B, their moduli may be highly similar, whereas their phases can differ significantly (e.g., $\phi_A = 0°$, $\phi_B = 120°$) [19].

4) Phase-Preserving Projection: To preserve phase information for subsequent CNN layers and the detection head, we employ a projection mechanism that first maps the phase to its sine and cosine components to avoid discontinuity, then concatenates them with the modulus and transforms the result to the real domain through a linear layer $\mathbf{W}_p$.

$$\mathbf{y}_t = \mathbf{W}_p\left([\,|\mathbf{h}_t^c|,\ \cos(\angle\mathbf{h}_t^c),\ \sin(\angle\mathbf{h}_t^c)\,]\right), \tag{4}$$

This design enables the detection head to more directly leverage phase-difference features, thereby effectively reducing feature entanglement among dense targets. Finally, the processed sequence is restored to a two-dimensional feature map $\mathbf{X}' \in \mathbb{R}^{C\times H\times W}$ through Inverse Snake Scan.

C. Design of the C2f_Mamba Module

C2f_Mamba is the core building block of YOLO26-Mamba3 and is specifically designed to replace the C2f modules at the P4 and P5 levels in the original YOLO26 neck network. This module adopts a dual-branch parallel structure that effectively fuses local CNN features with the global sequential context captured by Mamba3 while maintaining relatively lightweight computation.

As shown in Fig. 2, the input feature map $\mathbf{X}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ is first passed through a 1×1 convolution to compress the number of channels to $C' = \lfloor C \cdot e \rfloor$, where the expansion factor is $e = 0.5$. The compressed feature $\mathbf{X}_p$ is then split equally along the channel dimension into two branches, denoted as $\mathbf{X}_a$ and $\mathbf{X}_b$:

$$\mathbf{X}_p = \text{Conv}_{1\times1}(\mathbf{X}_{\text{in}}), \qquad [\mathbf{X}_a, \mathbf{X}_b] = \text{Split}(\mathbf{X}_p). \tag{5}$$

First, the local branch extracts texture and edge information by a lightweight depthwise convolution:

$$\mathbf{F}_a = \text{DWConv}_{1\times1}(\mathbf{X}_a). \tag{6}$$

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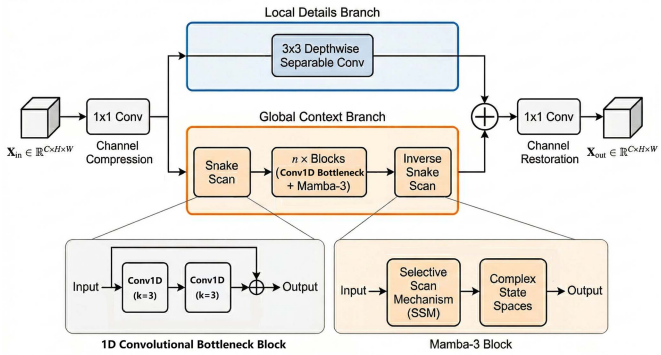


Fig. 2. Structure of the proposed C2f_Mamba module. The module adopts a parallel dual-branch design to effectively fuse local convolutional features with global sequence representations.

TABLE I
CONFIGURATION OF THE HYBRID NECK NETWORK

Level	Input Size	Module Type	Description
P3	80×80	C2f	Retain original CNN
P4	40×40	C2f_Mamba	Medium-scale global dependencies
P5	20×20	C2f_Mamba	Large-scale semantic context

Meanwhile, the global branch serializes $\mathbf{X}_b$ by Snake Scan and then processes the resulting sequence with n stacked layers, each composed of a 1D convolutional bottleneck and a Mamba3 block. The overall transformation can be expressed as:

$$\begin{aligned}\mathbf{S}^{(0)} &= \text{SnakeScan}(\mathbf{X}_b),\\ \mathbf{S}^{(i)} &= \text{Mamba3Block}\big(\text{Conv1D}_{k=3}(\mathbf{S}^{(i-1)})\big),\\ \mathbf{X}'_b &= \text{InverseSnakeScan}(\mathbf{S}^{(n)}).\end{aligned} \tag{7}$$

The features produced by the two branches are then concatenated along the channel dimension and fused by a 1×1 convolution to recover the original channel size, yielding the final output $\mathbf{X}_{\text{out}} \in \mathbb{R}^{C\times H\times W}$. Algorithm 1 summarizes the forward computation of the C2f_Mamba module.

D. Hybrid Neck Network

The neck of YOLO26 adopts a Feature Pyramid Network (FPN) structure that fuses multi-scale features through a top-down path and lateral connections. To enhance the model's discriminative ability in densely occluded pedestrian scenes while maintaining real-time performance, we replace the standard C2f modules at the low-resolution levels (P4 and P5) with the proposed C2f_Mamba module. The specific configuration is shown in Table I.

C2f_Mamba is introduced only at P4 and P5 to balance global modeling capability and computational efficiency. Because these low-resolution features contain richer semantics and shorter sequences, they are more suitable for Mamba-based long-range dependency modeling, whereas the high-resolution P3 feature mainly preserves fine-grained local details and is therefore kept as a lightweight CNN block. This design can be summarized as follows:

$$\mathcal{F}_{\text{neck}} = \{\,\text{C2f}(P3), \text{C2f_Mamba}(P4), \text{C2f_Mamba}(P5)\,\}, \tag{8}$$

where the assignment follows the basic principle that global sequence modeling is applied to low-resolution semantic features, while local convolution is still retained for high-resolution detail features.

In addition, introducing C2f_Mamba does not alter the original feature propagation path of the FPN. The upsampling and concatenation operations among P3, P4, and P5 remain unchanged, which helps preserve training stability and deployment compatibility.

Algorithm 1: Forward Computation of C2f_Mamba Module.

Require: Input feature map $\mathbf{X}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, number of stacked layers n, expansion factor e
Ensure: Output feature map $\mathbf{X}_{\text{out}} \in \mathbb{R}^{C\times H\times W}$

$\mathbf{X}_p \leftarrow \text{Conv}_{1\times1}(\mathbf{X}_{\text{in}})$ // Reduce channels
$\mathbf{X}_a, \mathbf{X}_b \leftarrow \text{Split}(\mathbf{X}_p)$ // Two branches
$\mathbf{F}_a \leftarrow \text{DWConv}_{1\times1}(\mathbf{X}_a)$ // Local branch
$\text{S}_{seq} \leftarrow \text{SnakeScan}(\mathbf{X}_b)$ // Serialize
for $i = 1$ to n **do**
  $\text{S}_{conv} \leftarrow \text{Conv1D}_{k=3}(\text{S}_{seq})$ // 1D convolution (kernel size 3) along sequence
  $\text{S}_{seq} \leftarrow \text{Mamba3Block}(\text{S}_{conv})$ // Update sequence
end for
$\mathbf{X}'_b \leftarrow \text{InverseSnakeScan}(\text{S}_{seq})$ // Restore 2D
$\mathbf{X}_{\text{cat}} \leftarrow \text{Concat}(\mathbf{F}_a, \mathbf{X}'_b)$
$\mathbf{X}_{\text{out}} \leftarrow \text{Conv}_{1\times1}(\mathbf{X}_{\text{cat}})$ // Restore channels
return $\mathbf{X}_{\text{out}}$

E. Loss Function

The total loss function consists of three terms:

$$\mathcal{L} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{rep}}\mathcal{L}_{\text{rep}}, \tag{9}$$

with balancing factors $\lambda_{\text{box}} = 7.5$, $\lambda_{\text{cls}} = 0.5$, and $\lambda_{\text{rep}} = 0.5$. $\mathcal{L}_{\text{box}}$ [20] is the Complete-IoU loss, which considers center distance and aspect ratio in addition to IoU:

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{g})}{c^2} + \alpha v, \tag{10}$$

where $\rho(\cdot)$ denotes the Euclidean distance, c is the diagonal length of the smallest enclosing box, v measures aspect-ratio consistency, and α is a trade-off parameter. $\mathcal{L}_{\text{cls}}$ [21] is the binary cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N}\sum_i \left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right], \tag{11}$$

where $y_i \in \{0, 1\}$ is the ground-truth label and p_i is the predicted probability. $\mathcal{L}_{\text{rep}}$ [22] penalizes predictions attracted to non-target pedestrians:

$$\mathcal{L}_{\text{rep}} = \frac{1}{N_+}\sum_{i\in\mathcal{P}}\sum_{j\in\mathcal{G}_{\text{rep}}(i)} -\ln(1 - \text{IoU}(\mathbf{b}_i, \mathbf{g}_j)), \tag{12}$$

where $\mathcal{P}$ is the set of positive predictions and $\mathcal{G}_{\text{rep}}(i)$ are ground-truth boxes not matched to $\mathbf{b}_i$ but with IoU > 0.1.

III. EXPERIMENTS

This section presents experiments conducted to evaluate the effectiveness of YOLO26-Mamba3 for dense pedestrian detection in railway scenes. Specifically, we first describe the

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TABLE II
COMPARISON RESULTS ON THE DENSE RAILWAY PEDESTRIAN DETECTION DATASET

Model	mAP@0.5:0.95 (%)	mAP@0.5 (%)	Recall@HighOcc (%)	OPD (%)	FPS
YOLOv8 [25]	48.3	73.1	41.2	32.5	57
YOLOv10 [26]	49.5	73.2	42.2	33.8	55
YOLOv11 [27]	51.4	76.4	44.7	36.1	52
YOLO26 [4]	54.3	79.2	48.5	40.3	64
Mamba-YOLO [28]	55.4	80.2	52.1	45.7	39
YOLO26+Transformer [29]	55.1	79.8	50.9	44.2	27
YOLO26-Mamba3(Ours)	**57.6**	**82.2**	**54.5**	**49.2**	42

TABLE III
ABLATION STUDY RESULTS ON THE DENSE RAILWAY PEDESTRIAN DETECTION DATASET

Configuration	mAP@0.5:0.95 (%)	Recall@HighOcc (%)	OPD (%)	Params (M)	FLOPs (G)
YOLO26	54.3	48.5	40.3	9.5	20.7
+ Serial Mamba3	55.8 (+1.5)	51.2 (+2.7)	44.1 (+3.8)	10.4	29.2
+ Complex state	56.9 (+1.1)	53.0 (+1.8)	47.5 (+3.4)	10.7	33.5
+ Repulsion loss	57.6 (+0.7)	54.5 (+1.3)	49.2 (+1.7)	10.8	37.6
YOLO26-Mamba3 (Ours)	**57.6**	**54.5**	**49.2**	10.8	37.6

dataset, evaluation metrics, and implementation details, then report quantitative comparison and ablation results, and finally provide qualitative visualization examples.

A. Experimental Setup

Dataset: Experiments are conducted on our self-built Dense Railway Pedestrian Detection Dataset, which contains 11,900 images in total, including 9,520 training images, 1,190 validation images, and 1,190 test images.

Evaluation metrics: We adopt standard COCO metrics, including mAP@0.5:0.95, mAP@0.5, and FPS [23]. We also use two occlusion-specific metrics. High-Occlusion Recall (Recall@HighOcc) measures recall on heavily occluded ground-truth pedestrians with a matching threshold of IoU > 0.5. Overlapping Pedestrian Detection (OPD) measures the proportion of overlapping pedestrian pairs that are correctly separated and matched to two distinct predictions [6].

Implementation Details: All experiments are conducted using the PyTorch 2.1 framework on two NVIDIA A100 GPUs. The AdamW optimizer [24] is employed with an initial learning rate of 0.01 and a cosine annealing scheduler. The total number of training epochs is 300.

B. Results on the Dense Railway Pedestrian Detection Dataset

Table II summarizes the comparison results of different detectors on the Dense Railway Pedestrian Detection Dataset. YOLOv8 and YOLOv11 provide high inference speed, but their accuracy and high-occlusion recall remain limited in crowded railway scenes. YOLO26 improves both accuracy and efficiency over these conventional baselines, while Mamba-YOLO further enhances occlusion-related performance at the cost of lower speed. In contrast, YOLO26+Transformer suffers from a clear efficiency drop, whereas the proposed YOLO26-Mamba3 achieves the best overall results in mAP, Recall@HighOcc, and OPD while maintaining a practical model size and inference speed, demonstrating a better trade-off between detection accuracy and real-time deployment.

C. Ablation Studies

As shown in Table III, compared with the baseline YOLO26, the proposed YOLO26-Mamba3 improves mAP@0.5:0.95 from 54.3% to 57.6%, Recall@HighOcc from 48.5% to 54.5%, and OPD from 40.3% to 49.2%, corresponding to gains of 3.3, 6.0, and 8.9 percentage points, respectively. These results show that the integration of serialized Mamba3, complex-state modeling, and repulsion loss substantially enhances dense pedestrian detection, particularly in heavily occluded scenes where distinguishing adjacent individuals is critical. The consistent improvements across all three metrics further confirm that the proposed design strengthens both overall detection accuracy and instance-separation ability in dense railway environments.



Fig. 3. Representative samples from the Dense Railway Pedestrian Detection Dataset.

D. Qualitative Visualization

To provide a qualitative overview of the Dense Railway Pedestrian Detection Dataset used in our experiments, a representative training batch is shown in Fig. 3. The figure illustrates high pedestrian density, large-scale variation, and frequent mutual occlusion in railway scenes. The detection results further demonstrate the effectiveness of our method.

IV. CONCLUSION

In this paper, we propose YOLO26-Mamba3, a hybrid architecture that integrates YOLO26 with Mamba3 through selective state space and complex state space. The architecture retains the anchor-free and NMS-free inference pipeline and introduces a serialized Mamba3 module into the neck network. The Selective SSM dynamically focuses on discriminative features under occlusion, while the complex state space encodes relative spatial relationships as phase differences to distinguish heavily overlapping pedestrians, further assisted by a repulsion loss. Experimental results demonstrate that YOLO26-Mamba3 significantly outperforms the baseline YOLO26 under highly occluded conditions. We acknowledge that the current evaluation is limited to our railway-specific dataset, and generalization to public dense-crowd benchmarks remains to be explored in future work.

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